# Timo Probst

Research Assistant
Chair of Computer Graphics, Department for Computer Science
University of Freiburg, Germany

probstt@informatik.uni-freiburg.de   Personal Website   LinkedIn   Google Scholar

## Education

**Ph.D. Student** — University of Freiburg, Germany
*Chair of Computer Graphics, Department of Computer Science* — *2019 – 2021*
*Supervisor: Matthias Teschner*

**M.Sc. in Computer Science** — University of Freiburg, Germany
*Graduated with honors.* — *2019 – 2021*
*Thesis title: Global Contact Handling and Dry Friction for Particle-based Rigid Bodies.*

**B.Sc. in Computer Science** — University of Freiburg, Germany
*Thesis title: Matrix-free PPE Solver for SPH Fluids* — *2015 – 2019*

## Employment

**Research Assistant** — University of Freiburg, Germany
*Chair of Computer Graphics, Department of Computer Science* — *2021 – ...*

**Student Assistant** — University of Freiburg, Germany
*Chair of Computer Graphics, Department of Computer Science* — *2019 – 2021*

**Tutor** — University of Freiburg, Germany
*Python for Biologists, Faculty of Biology* — *2019*

## Publications

[1] Timo Probst and Matthias Teschner. “Unified Pressure, Surface Tension and Friction for SPH Fluids”. In: *ACM Trans. Graph.* 44.1 (Dec. 2024). ISSN: 0730-0301. DOI: 10.1145/3708034.

[2] Timo Probst and Matthias Teschner. “Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH”. In: *Computer Graphics Forum* 42.1 (2023), pp. 155–179. DOI: 10.1111/cgf.14727.

## Invited Talks

**SIGGRAPH 2025** — Vancouver, Canada
*Unified Pressure, Surface Tension and Friction for SPH Fluids* — *August 2025*

**FIFTY2** — Freiburg, Germany
*Simulating Water Droplets with SPH* — *December 2024*
*Friction and Contact Handling for Rigid Bodies and Fluids Using SPH* — *March 2023*

**Eurographics 2023** — Saarbrücken, Germany
*Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH* — *May 2023*

## Reviews

ACM SIGGRAPH — *2025, 2026*

ACM SIGGRAPH Asia — *2026*

IEEE Transactions on Visualization and Computer Graphics — *2023, 2025, 2026*

Eurographics — *2026*

ACM SIGGRAPH / Eurographics Symposium on Computer Animation — *2026*

Journal of Computational Physics — *2025, 2026*

Nuclear Engineering and Design — *2025*

## Grants

**Deutschlandstipendium** — Germany
*Selected based on academic performance and social commitment.* — *2019 – 2020*

## Supervision

### Graduate Students

Julian Karrer — University of Freiburg
*2026*

Joshua Fässler — University of Freiburg
*Strong Rigid-Fluid Coupling using a Unified Implicit SPH Solver* — *2024 – 2025*

Philipp Sandweg — University of Freiburg
*Implicit SPH Simulation of Elastic Solids in a Multi-Material Framework* — *2023 – 2024*

Andrés Sandoval — University of Freiburg
*Global Matrix-free Pressure Solver for Performant SPH Fluid Simulations* — *2022 – 2023*

### Undergraduate Students

Johann-Frederik Isele — University of Freiburg
*2026*

Marius Hägele — University of Freiburg
*Implementation and evaluation of a 2D state equation SPH fluid solver* — *2025 – 2026*

Georgij Hergenräder — University of Freiburg
*Smoothed Particle Hydrodynamics Fluid Simulation in 2D - Implementation and Evaluation* — *2025 – 2026*

## Skills

**Languages**: Strong reading, writing and speaking competencies for English, German
**Coding**: C++, Python, C, C#, OpenCL, SYCL, CUDA
**Frameworks**: NumPy, Godot, Unreal Engine, Blender